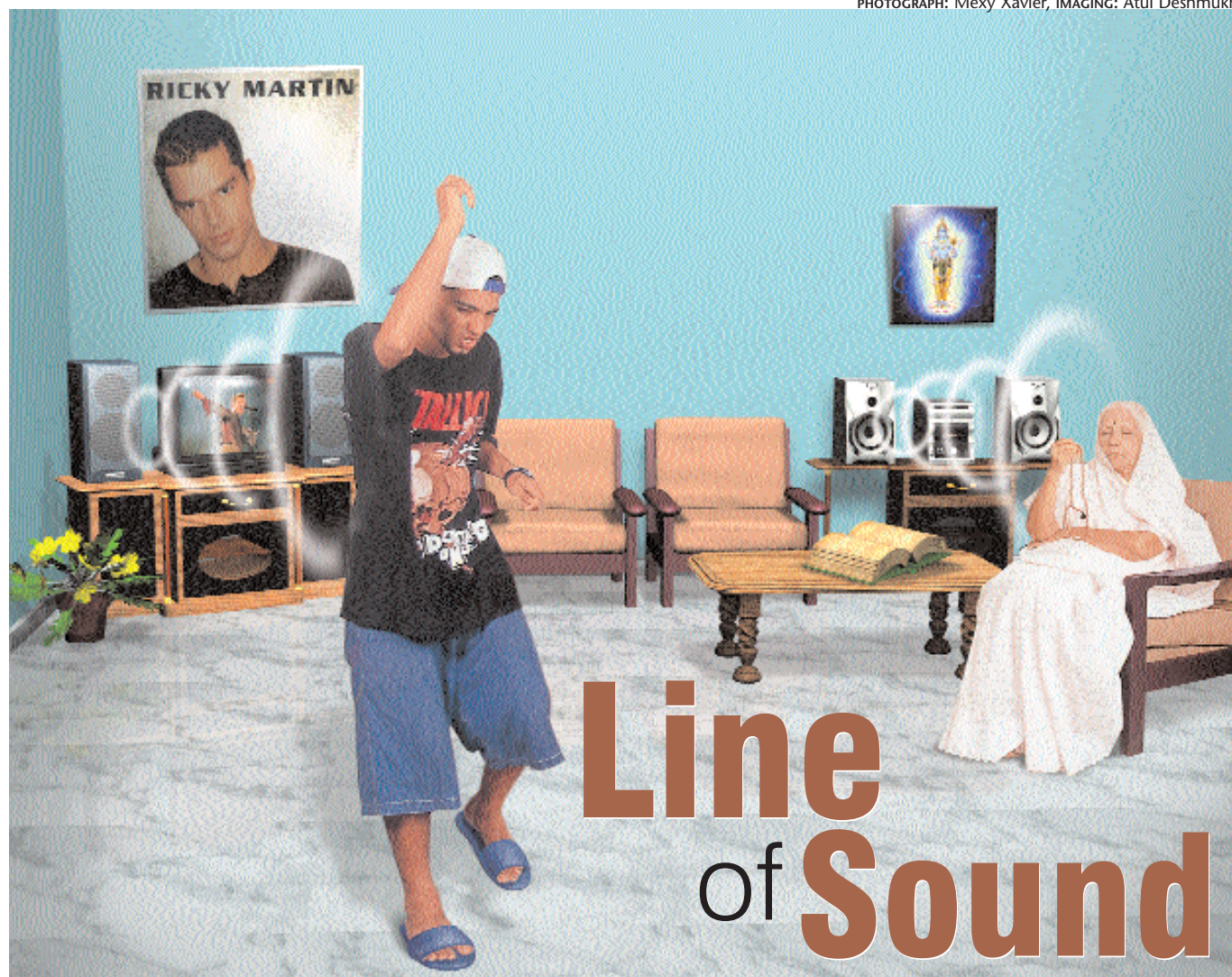




Line of Sound

Sound, like light can now be directed, so that only those within its path can hear it.

For many decades now, the way we listen to music hasn't really changed. If you claim to be an audiophile, or a connoisseur of fine music, then you probably have your own personal acoustical preferences in instruments that reproduce music. However, it's interesting to know that the basic technology powering the speaker has remained just about the same.

At the moment, two companies that have been at the forefront of research in acoustics have decided to change that. The secret weapon in their arsenal is an often touted, but never quite realised technology that creates focused beams of sound, similar to the light beams coming out of a flashlight. The technology that powers this is known as audio spotlighting. It uses a combination of non-linear acoustics and some very fancy and com-

plex mathematics. But it's real, and works fine enough to knock the socks of any other conventional loudspeaker.

Researchers have worked for many years to focus sound into a coherent and highly directional beam. It's never quite worked beyond the laboratory; at least, not until recently, when former MIT Media Labs researcher, Joseph Pompei, managed to crack the problem by studying current techniques and re-engineering them. Now, Holosonic Research Labs and the American Technology Corporation have come up with audio solutions that make use of audio spotlighting technology to create fine audio solutions for a wide variety of applications.

Lighting up sound

Audio spotlighting makes for an interesting theory, but you just can't focus, or

direct a beam of sound the way you focus light using a flashlight. Perhaps just a little, by holding an open-mouthed funnel-shaped pipe close to your mouth and shouting. However, this still doesn't direct the sound as required.

The audible portions of sound tend to spread out in all directions from the point of origin. They do not travel as narrow beams—which is why you don't need to be right in front of a radio to hear music. In fact, the beam angle of audible sound is very wide, just about 360 degrees. This effectively means the sound that you hear will be propagated through air equally, in all directions.

In order to focus sound into a narrow beam, you need to maintain a low beam angle that is dictated by wavelength. The smaller the wavelength, the less the beam angle, and hence, the more